Scholar No:..... Scholar Name:.....

MAULANA AZAD NATIONAL INSTITUTE OF TECHNOLOGY, BHOPAL
DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
MIDTERM EXAMINATION (November 2024)

Course: M.Tech. Semester: I Date:23/11/2024
Sub: Web Search and Information Retrieval Max. Marks: 20
Time: 1 hour Sub Code: FAC-560/FCN 558/ FIS 561/ FAI 558
Branch: All Branches

Note 1: Attempt All Questions.

Q.No.	Questions	Marks	Cos
1.	Consider the following documents: **Doc1:** "The skilled engineer designs a robust system." **Doc2:** "A talented programmer writes efficient code." **Doc3:** "The experienced manager leads a productive team." a) Construct the inverted index. Assume stop words (e.g., "the," "a," "an") are removed, and all terms are converted to lowercase. b) Determine which documents are retrieved for the following queries: **Query 1:** "engineer AND system" **Query 2:** "team AND NOT productive"	05	CO1
2.	Discuss the importance of evaluating an information retrieval system and describe the key performance metrics used for the evaluation. Two information retrieval systems, IRX and IRY, are tested on the same query over a collection of 500,000 documents. IRX retrieved 40,000 documents, out of which 6,000 are relevant, while IRY retrieved 50,000 documents, out of which 4,000 are relevant. Assuming the total number of relevant documents in the collection is 10,000: a) Compute **Precision** and **Recall** for both systems. b) Identify which system performs better based on these metrics and justify your conclusion.	05	CO2
3.	Consider the following set of words in a vocabulary: Words: teacher, doctor, engineer, artist, scientist a) Construct the permuterm index for the given words. b) Using the constructed permuterm index, determine which words would be retrieved for the following queries: **Query 1:** *ist **Query 2:** eng* **Query 3:** t*er	05	CO2
4.	The Soundex algorithm is used to encode words into phonetic representations based on their pronunciation. Consider the following list of names: **Robert, Rupert, Ashcraft, Pfister, Honeyman** a) Calculate the Soundex code for each name in the list. b) Find the Soundex matches for the following names: **Query 1:** Rupert **Query 2:** Ashcroft	05	CO3

Department of Computer Science and Engineering

End Term Exam (Dec - 2024)

M. Tech. First Semester — **Subject: Computer and Network Security**

Sub Code: FIS 512/FCN 513 — Time: 02 Hr. Max. Marks: 50

Note: All questions are compulsory to attempt. Answer should be laconic. Make suitable assumptions wherever necessary and clearly state the same.

Q-1 A What are the potential limitations of signature-based IDS? Write SNORT rules to perform the following intrusion detection task — CO1 — 05

I. Detecting a Specific Payload for example DNS query detected
II. Detecting SQL Injection via "1=1"
III. Detecting SYN Scan (TCP Port Scan)

B Use the following firewall policy rules, with a "default DENY" default rule. — CO3 — 05

Rule #	Target	Protocol	Source	Destination
1	ALLOW	udp	*:*/out	23.45.68.5:53/in
2	ALLOW	tcp	*:*/out	23.45.68.10:80/in
3	ALLOW	tcp	23.45.68.10:80/in	*:*/out
4	ALLOW	tcp	*:*/out	23.45.68.11:80/in
5	DROP	all	*:*/out	23.45.68.*:*/in
6	ALLOW	tcp	23.45.67.*:*/out	23.45.68.*:*/in
7	ALLOW	udp	23.45.67.3:8000/out	23.45.68.3:4058/in
8	DROP	tcp	128.59.*.*:*/out	23.45.68.5:*/in
9	ALLOW	tcp	128.59.17.*:*/out	23.45.68.*:1099/in

I. Identify any redundant or conflicting parts of this firewall policy.
II. For each of the following packets, state which rule applies and how the packet would be handled by the firewall.
   a. TCP 66.68.80.90:7800 → 23.45.68.10:80
   b. UDP 66.68.80.90:9300 → 23.45.68.5:53
   c. UDP 128.59.16.2:5589 → 23.45.68.3:4058

Q-2 A What are the main differences between IPSec's transport mode and tunnel mode. An IPSec tunnel uses ESP with the following overheads: — CO1 — 05
ESP Header: 20 bytes
ESP Authentication: 16 bytes
ESP Trailer (Padding): 12 bytes
If the original packet size is 1,200 bytes, what is the total size of the packet after IPSec processing?

B Explain the Kerberos authentication process, starting from when a user logs in to a system, and including the interactions between the client, KDC, and the service the client wants to access. What are some potential security vulnerabilities in the Kerberos protocol — CO3 — 05

Q.3 A	What are the fields in an X.509 certificate, and what are their roles?	CO1	04
B	Why is the Smurf Attack considered a Distributed Denial of Service (DDoS) attack, and what makes it different from other DoS attacks? Suppose an attacker launches a Smurf attack on a target with an IP address of 10.0.0.10. The attacker uses a network with the following characteristics: • The broadcast address for the network is 10.0.0.255. • The network consists of 200 devices that respond to each ICMP Echo Request. • The size of each ICMP Echo Request is 80 bytes, and the size of each ICMP Echo Reply is 80 bytes. Calculate: I. The total amount of traffic the victim will receive. II. The amplification factor for this attack.	CO3	06
Q.4 A	What is an IP Fragmentation Attack (Teardrop ICMP/UDP)? Suppose we have to transmit a datagram of size 4000 bytes from source node to destination node with the Maximum Transmission Unit (MTU) of intermediate routers is 1000 bytes. Then calculate the number of fragments and value of fragmentation offset of each IP fragment. (The Size of IP Header is 20 bytes).	CO2	06
B	How could an attacker exploit Cross Site Scripting vulnerability to trigger the execution of malicious code?	CO2	04
Q.5 A	YouTube employs various technologies to safeguard the content uploaded by users, with Digital Rights Management (DRM) being one of the primary methods. Describe how DRM is used on YouTube to prevent unauthorized downloading, distribution, and piracy of copyrighted videos. Highlight the main DRM techniques used, assess their effectiveness, and explore the challenges YouTube faces in balancing content protection with the user experience.	CO2	06
B	How taint propagation using static analysis is used to detect SQL. Give example	CO2	04

Name of the Student: Riya Kumbhare Roll No.: 24212011113

MAULANA AZAD NATIONAL INSTITUTE OF TECHNOLOGY, BHOPAL

HUMANITIES AND SOCIAL SCIENCES

EXAMINATION: MID TERM **MONTH & YEAR** : March 2025

Course: M Tech Group C (Open Elective) **Semester:** II

Subject Name: Intellectual Property Rights For Engineers **Sub Code: HUM-581**

Note : Answer All Questions.

Q.No.	Questions	Marks
1. a	Define creativity and novelty? How does creativity leads to invention?	2.5
1.b	What are the basic requisites of patentability criteria?	2.5
2.	What is geographical indication? What rights does a geographical indication provides? List five items of India that are granted with "geographical indications" tag. Mention at least two from Madhya Pradesh.	5
3.	"Internet poses new challenges to copyright rights protection in the digital era". Enumerate the statement.	5
4.	You are in the process of picking a name for your new business. What should you keep in mind to pick a good and protectable name?	5

Name of the Student..... Scholar Number.....

MAULANA AZAD NATIONAL INSTITUTE OF TECHNOLOGY BHOPAL
DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

EXAMINATION: MID TERM **MONTH and YEAR: March 2025**

Course: M.Tech. (CN & AI) **Semester:** II **Subject Code:** FAI 579
Subject Name: Natural Language Processing **Maximum Marks:** 20
Duration: 1 hr. **Date:** 19/03/2025 **Time:** 11:30 am to 12:30 pm

Note: *All questions are compulsory.*

Q. No.	Questions	Marks	COs
01	(i) Given the string: "The price is $100.50, and the discount is 20%.", write a regular expression to extract: (a) all monetary values (e.g., $100.50) and (b) all percentages (e.g., 20%) a-) \$\d+ (\.\d{2})? b-) \d+% (ii) Given the sentence: "The bank is closed.", explain the potential ambiguities and how they can be resolved using context. (iii) Given the following corpus "the cat sat on the mat" "the dog sat on the log" "the cat and the dog are friends" Compute the probability of the sentence "the cat sat on the log" and perplexity using trigram model (iv) 'Time flies like an arrow', draw the parse tree to prove that the sentence has ambiguity	05	CO1 CO3
02	Find the Levenshtein edit distance between "elephant" and "relevant". Backtrack and show the operations performed.	03	CO1
03	Consider the following data: "she sells seashells by the seashore she sells seashells near the shore" Find the probability of the sentence "sells seashells by the shore", using Good-Turing discounting algorithm	05	CO2 CO3
04	Derive a top-down, depth-first, left-to-right parse tree for the given sentence: *The angry bear chased the frightened little squirrel* Use the following grammar rules to create the parse tree: S → NP VP NP → Det Nom VP → V NP Nom →Adj Nom | N Det → the Adj→ little | angry | frightened N → squirrel | bear V → chased	03	CO2 CO3
05	Apply CKY algorithm to check whether the string "A pilot likes flying planes" is accepted by given grammar or not. S → NP VP NP → DT NN NP →JJ NNS VP → VBG NNS VP → VBZ VP VP→ VBZ NP DT → A NN → pilot VBZ → likes VBG → flying JJ → flying NNS → planes	04	CO2 CO3

**********END**********

Name of the Student Nurshan Ajmeri 2321201102 Roll No.

MAULANA AZAD NATIONAL INSTITUTE OF TECHNOLOGY

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

EXAMINATION: ENDTERM EXAM **MONTH & YEAR:** April-May 2024

Course: M. Tech **Semester:** II **Branch:** CN

Subject Name: Graph Theory and Network Algorithms **Subject Code:** CN 203

Time: 180 minutes **Max Marks:** 50

NOTE: Answer all Questions. Attempt all the parts of a Question in continuation.

Q.1(a) The complete m-partite graph $K_{n1, n2, \ldots, nm}$ has vertices partitioned into m subsets of $n_1, n_2, \ldots, n_m$ elements each, and vertices are adjacent if and only if they are in different subsets in the partition.

(i) How many vertices and edges does the complete m-partite graph $K_{n1, n2, \ldots, nm}$ have? (5)(CO1)

(ii) How many paths of length '4' exists in $K_{6, 7, 8}$?

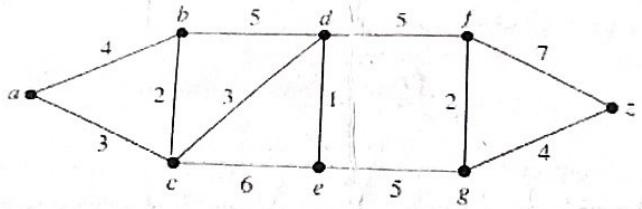
(iii) How many cycles of length '4' exists in $K_{6, 7, 8}$?

(b) Let $k \in Z+$, $k \geq 3$. If $G = (V, E)$ is a connected planar graph and each cycle of length at least k, prove that no. of regions R in the planar graph is

$$R \leq \frac{2(|V|-2)}{(k-2)}$$ (3)(CO2)

(c) Let G(V, E) be a connected undirected bipartite graph with V partitioned as $V1 \cup V2$. State the necessary and sufficient condition to have a Hamilton cycle and path in G. Justify your answer. (2)(CO1)

Q.2(a) Find the shortest path from vertex 'a' to 'z' using Dijkastra Algorithm in the following graph G.(5)(CO3)



(b) The wheel graph, W_n, is obtained from a cycle of length n by placing a new' vertex within the cycle and adding edges (spokes) from tills new vertex to each vertex of the cycle. Prove using mathematical induction that chromatic polynomial P is

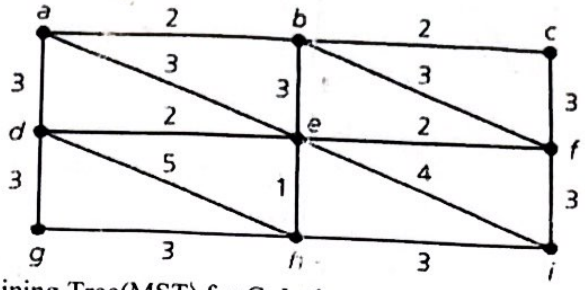
$$P(W_n, \lambda) = \lambda(\lambda-2)^n + (-1)^n\lambda(\lambda-2)$$ (5)(CO2)

Q.3(a) Find all (loop-free) non-isomorphic connected undirected graphs with four vertices. (2)(CO2)

(b) Consider the following subjects: CSE 115, CSE 116, CSE 185, CSE 195, CSE 101, CSE 102, CSE 273, and CSE 473. Schedule the final exams using the fewest number of different time slots, if there are no students taking both CSE 115 and CSE 473, both CSE 116 and CSE 473, both CSE 195 and CSE 101, both CSE 195 and CSE 102, both CSE 115 and CSE 116, both CSE 115 and CSE 185, and both CSE 185 and CSE 195, but there are students in every other pair of courses. (4)(CO2)

Name of the Student.....Rija

(c) Consider the following graph G: (4)(CO3)



Find the Minimum Spaining Tree(MST) for G. Is there more than one MST of G.

Q.4(a) A telephone communication system is set up at a company where 125 executives are employed. The system is initialized by the president, who calls her four vice presidents. Each vice president then calls four other executives, some of whom in turn call four others, and so on. (Each executive who does make a call will actually make four calls.) (3)(CO3)

(i) How many calls are made in reaching all 125 executives?

(ii) How many executives, aside from the president, are required to make calls?

(b) Let T be a binary tree with the vertices of T ordered by a preorder traversal. This traversal assigns the label 1 to all internal vertices of T and the label 0 to each leaf. The sequence of 0's and 1's that results from the preorder traversal of T is called the tree's characteristic sequence. Determine the binary trees for the characteristic sequences: (3)(CO3)

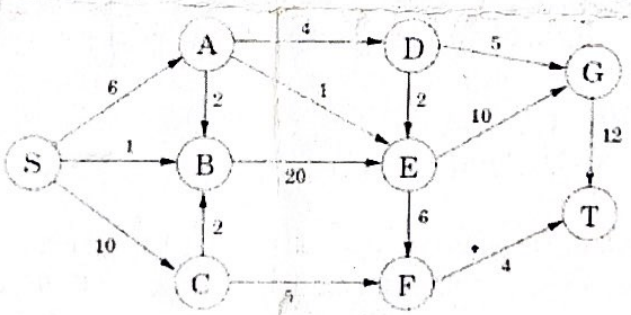
(i)1011110000101011000 (ii) 11101100110010101100

(c) A mathematical equation supports three binary operators: #, $, & where # has highest and & has least precedence amongst them. The associativity of # and & is left while $ has right associative. (4)(CO3)

Construct expression tree for the following expressions:

(i) a#b#cde&f&g$h (ii) a$b$c#d$e&f#g#h&i$j

Q. 5 (a) Consider the following graph G:



Find the max flow supported by G. (5)(CO4)

(b) Draw the network diagram for the following project. Find the critical path and maximum time for completion of the project. (5)(CO4)

Activity	A	B	C	D	E	F	G	H	I	J	K	L
Preceded by	-	-	A	B	D	D, C	C	F	D	G, H, F	E	I
Duration (weeks)	10	9	7	6	12	6	8	8	4	10	5	8

Name of the Student.....Riya Scholar Number 24213011113

Name of the Student..... Scholar Number.....23212

MAULANA AZAD NATIONAL INSTITUTE OF TECHNOLOGY BHOPAL

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

EXAMINATION: MID TERM **MONTH and YEAR: MARCH 2024**

Course: M.Tech **Semester: II** **Branch: Computer Networking** **Subject Code: CN 523**

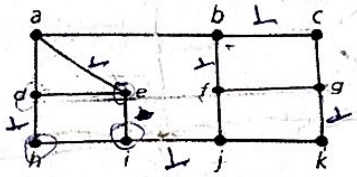
Subject Name: Graph Theory and Network Algorithms **Maximum Marks: 20**

Duration: 90 Minutes **Date: 21/03/24** **Time: 09:30 AM to 11:00 AM**

Note: (i) Attempt all the Questions. All the Questions carry equal Marks.
(ii) Attempt all the part of a question in continuation

Q 1. For $n \in N$, the n-dimensional hypercube (or n-cube) is denoted by Q_n. It is a loop-free connected undirected graph with 2^n vertices. For $n \geq 1$, these vertices are labeled by the 2^n n-bit sequences representing 0, 1, 2, ..., $2^n - 1$. Two vertices v_1 and v_2 of Q_n are joined by the edge (v_1, v_2) when the binary labels for v_1, v_2 differ in exactly one position. Find the number of edges in Q_n. **(2) CO1**

Q 2. The following graph shows an undirected graph representing a section of a department store. The vertices indicate where cashiers are located; the edges denote unblocked aisles between cashiers. The department store wants to set up a security system where (plainclothes) guards are placed at certain cashier locations so that each cashier either has a guard at his or her location or is only one aisle away from a cashier who has a guard. What is the smallest number of guards needed? **(2) CO1**

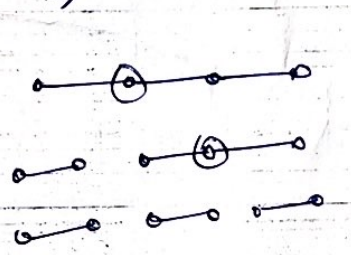


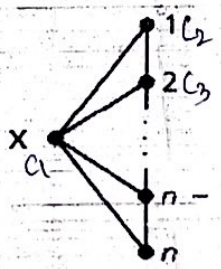
Q 3. Let T = (V, E) be a tree with $|V| = n > 3$. **(4) CO3**

(a) What are the smallest and the largest numbers of articulation points that T can have? Describe the trees for each of these cases.

(b) How many biconnected components does T have in each of the cases in part (a)?

Q 4. Find the chromatic number of the following Graph G **(4) CO2**



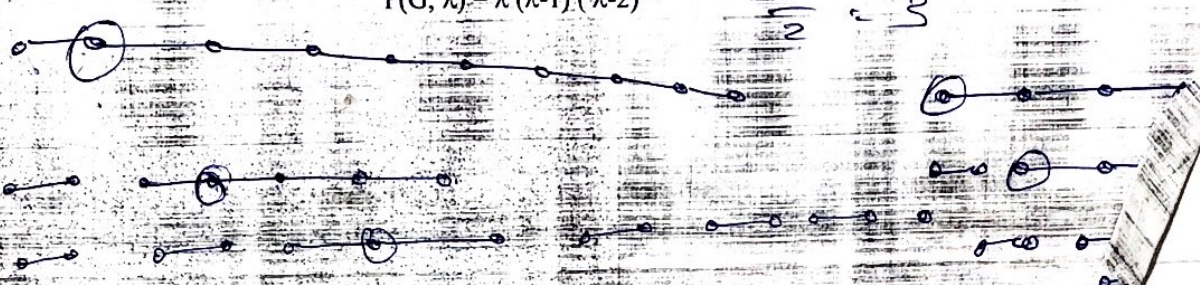




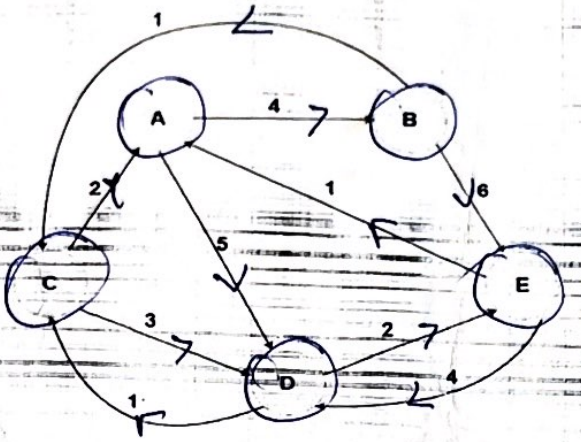
Prove using Mathematical Induction that Chromatic Polynomial

$$P(G, \lambda) = \lambda (\lambda-1) (\lambda-2)^{n-1}$$

PTO



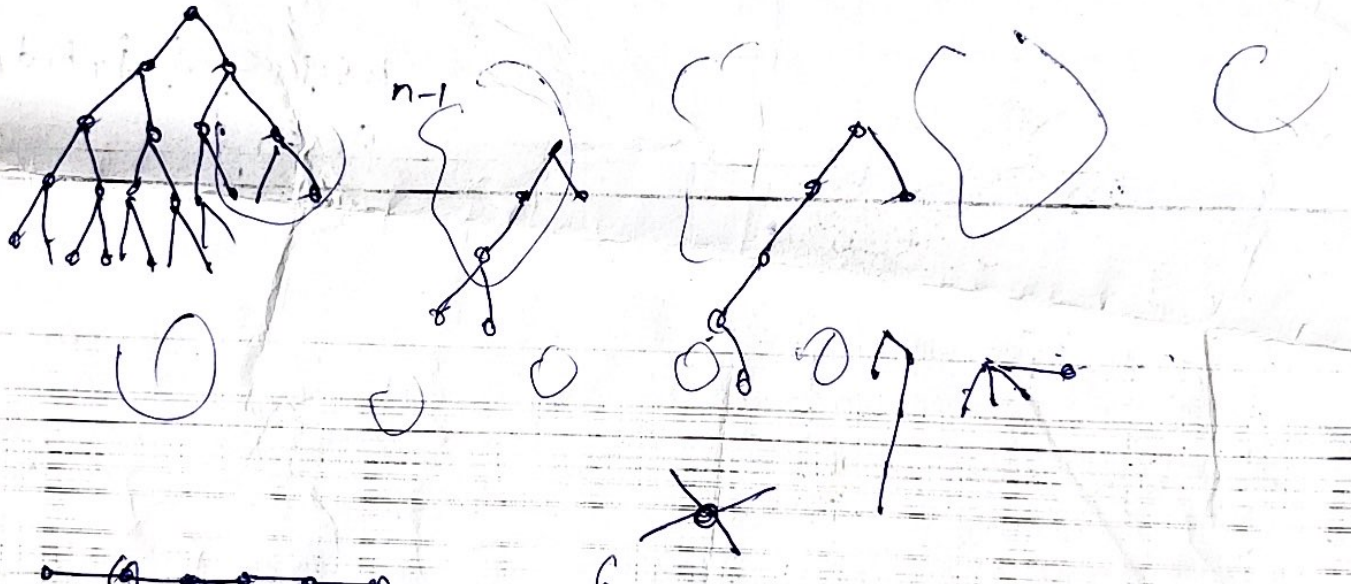
5. Consider the following graph G(V, E, W): **(4) CO2**

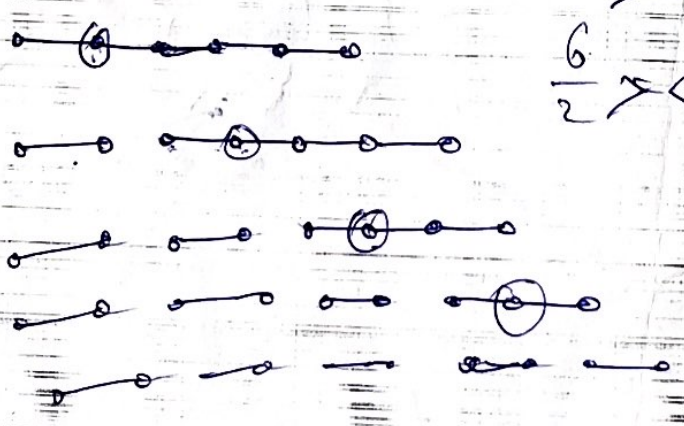


Find all pair shortest path.

Q 6. (a) Let F (V, E) be a forest of 10 trees where |E| = 40. What is |V|? **(4) CO3**

(b) If F (V, E) is a forest with |V| = m and |E| = n, how many trees determine F?





$\frac{6}{2} > 3+1 =$

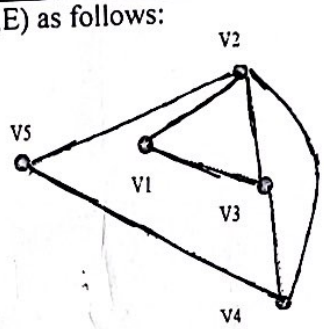
Name of the Student.....Riya..... Scholar Number.....2421201113

MAULANA AZAD NATIONAL INSTITUTE OF TECHNOLOGY BHOPAL
DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

EXAMINATION: MID TERM **MONTH and YEAR: March 2025**

Course: M.Tech. (CN) Semester: 2nd **Subject Code: FCN24522**
Subject Name: Graph Theory and Network Algorithms **Maximum Marks:** 20
Duration: 1 Hour **Date:** 18/03/2025 **Time:** 3:30PM to 4:30 PM

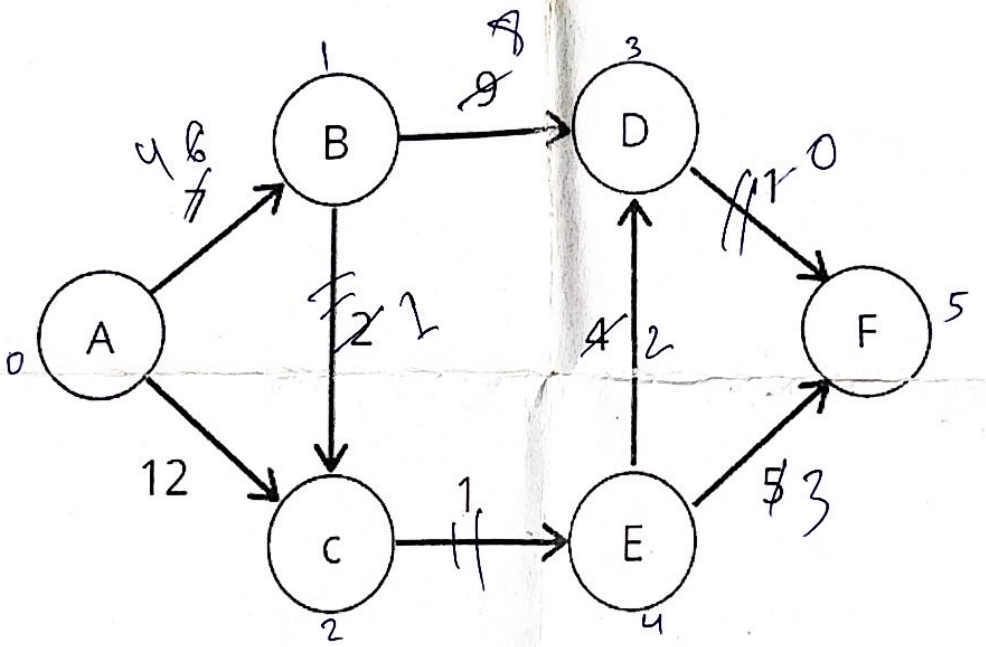
Note: *Attempt all the questions. Make any necessary and suitable assumptions, if required.*

Q. No.	Questions	Marks	COs
1 (a)	Let T = (V, E) be a tree with $\|V\| = n > 3$. What are the smallest and the largest numbers of articulation points that T can have? Describe the trees for each of these cases.	04	CO3
(b)	How many bi-connected components does T have in each of the cases in part (a)?		
2	Prove that if G be a graph of order $p \geq 3$ and deg $v \geq p/2$ for every vertex of G, Them G is Hamiltonian.	06	CO2
3	Consider a graph G(V,E) as follows: 		CO1
(a)	Find the Chromatic polynomial of the graph G.	04	
(b)	How many Paths of length 2 in G.	02	
4 (a)	Schedule the final exams for Math 115, Math 116, Math 185, Math 195, CS101, CS 102, CS 273, and CS 473, using the fewest number of different time slots, if there are no students taking both Math 115 and CS 473, both Math 116 and CS 473, both Math 195 and CS 101, both Math 195 and CS 102, both Math 115 and Math 116, both Math 115 and Math 185, and both Math 185 and Math 195, but there are students in every other pair of courses.	04	CO2

**********END**********

Graph Theory and Algorithms LAB

1. Write a program to find the Single Source Shortest Path using the dijkstra algorithm, use the example case of graph gien below(A is source and F is destination node).
2. Write a program to find the minmum spanning tree using prims algorithm, use the example case of graph gien below.



मौलाना आज़ाद राष्ट्रीय प्रौद्योगिकी संस्थान, भोपाल – ४६२००३
संगणक विज्ञान एवं अभियांत्रिकी विभाग
Maulana Azad National Institute of Technology, Bhopal – 462003
DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

Mid Term Examination, March 2024

Programme: **M. Tech/Ph.D** | Semester: **II**
Subject: **Research Methodology** | Subject Code: **MTH -524**
Branch: **AI/AC/CN/IS** | Max Marks: **20**
Date: **March 20, 2024** | Exam Time: **03:30PM to 05:00PM**

S.N		Question	Mark
1.	I.	"A research scholar has to work as a judge and derive the truth and not as a leader who is only eager to prove his case in favour of his plaintiff." Discuss the statement pointing out the objectives of research.	2+2
	II.	Discuss interview as a technique of data collection.	
2.	I.	Explain the need of cross validation and their types while evaluating estimator performance.	2+2
	II.	Suppose that we want to select between two prediction models, M1 and M2. We have performed 10 rounds of 10-fold cross-validation on each model, where the same data partitioning in round *i* is used for both M1 and M2. The error rates obtained for M1 are 30.5, 32.2, 20.7, 20.6, 31.0, 41.0, 27.7, 26.0, 21.5, 26.0. The error rates for M2 are 22.4, 14.5, 22.4, 19.6, 20.7, 20.4, 22.1, 19.4, 16.2, 35.0. Comment on whether one model is significantly better than the other considering a significance level of 1% (use the table 3.2).	
3.		Write short notes on the following:	2+2
	I.	The techniques of writing report	
	II.	Characteristics of a good research report	
4.	I.	Two hundred bolts were selected at random from the output of each of the five machines. The number of defective bolts found were 5, 9, 13, 7 and 6 .	3+2

2321201103

$\frac{1}{6} \cdot 139$

5-1=4

Is there a significant difference among the machines? Use 5% level of significance (refer the table 3.2).

II. Clearly explain the difference between collection of data through questionnaires and schedules.

5. From the data given in the following table 3.1, find out whether there is any relationship between gender and the preference of colour given 5 level of significance (use the table 3.2):

Table 3.1

Colour	Male	Female	Total
Red	25	45	70
Blue	45	25	70
Green	50	10	60
Total	120	80	200

Table 3.2

d.f.	.995	.99	.975	.95	.9	.1	.05	.025	.01
1	0.00	0.00	0.00	0.00	0.02	2.71	3.84	5.02	6.63
2	0.01	0.02	0.05	0.10	0.21	4.61	5.99	7.38	9.21
3	0.07	0.11	0.22	0.35	0.58	6.25	7.81	9.35	11.34
4	0.21	0.30	0.48	0.71	1.06	7.78	9.49	11.14	13.28
5	0.41	0.55	0.83	1.15	1.61	9.24	11.07	12.83	15.09
6	0.68	0.87	1.24	1.64	2.20	10.64	12.59	14.45	16.81
7	0.99	1.24	1.69	2.17	2.83	12.02	14.07	16.01	18.48
8	1.34	1.65	2.18	2.73	3.49	13.36	15.51	17.53	20.09
9	1.73	2.09	2.70	3.33	4.17	14.68	16.92	19.02	21.67
10	2.16	2.56	3.25	3.94	4.87	15.99	18.31	20.48	23.21
11	2.60	3.05	3.82	4.57	5.58	17.28	19.68	21.92	24.72
12	3.07	3.57	4.40	5.23	6.30	18.55	21.03	23.34	26.22

**********Best_of_Luck**********

P321201103

मौलाना आज़ाद राष्ट्रीय प्रौद्योगिकी संस्थान, भोपाल – ४६२००३
संगणक विज्ञान एवं अभियांत्रिकी विभाग

Maulana Azad National Institute of Technology, Bhopal - 462003 DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

End Semester Examination, April-May 2024

Programme: **M. Tech/Ph.D** Semester: **II**

Subject: **Research Methodology** Subject Code: **MTH -524**

Branch: **AI/AC/CN/IS** Max Marks: **50**

Date: **May 4, 2024** Exam Time: **09:00AM to 12:00PM**

S.N	Question	Mark	COs
1.	a) Imagine you're embarking on a research project to investigate the impact of social media on teenagers' mental health. Explain the foundational elements you would consider before initiating your research.	4+3+3	CO1 CO2
	b) Explain the process of problem identification and formulation in research methodology, highlighting its importance and key steps.		
	c) Discuss the qualities of a good hypothesis in research and explain why each quality is important		
2.	a) Research design in exploratory studies must be flexible but in descriptive studies, it must minimize bias and maximize reliability. Discuss.	4+3+3	CO1 CO2
	b) How does the case study method differ from the survey method? Analyze the merits and limitations of case study method in sociological research.		
	c) Differentiate between research hypotheses and null hypotheses and illustrate your answer with suitable example.		
3.	a) Historical researches may have hypotheses but these are different from the hypotheses of scientific researchers. Comment on this statement.	3+4+3	CO1 CO3
	b) A die is thrown 132 times with following results:		

Number	1	2	3	4	5	6
Frequency	16	20	25	14	29	28

Perform X^2 test and check Is the die unbiased? Where 5 per cent level of significance is 11.071

c) Why are research ethics important for researchers and academics? Explain various research ethics issues specially for research publication.

4. a) Genetic theory states that children having one parent of blood type A and the other of blood type B will always be of one of three types, A, AB, B and that the proportion of three types will on an average be as 1 : 2 : 1. A report states that out of 300 children having one A parent and B parent, 30 per cent were found to be types A, 45 per cent per cent type AB and remainder type B. Test the hypothesis by X^2 test. Use: at 5 per cent level of significance is 5.991. 5+5 CO1 CO2 CO3

b) Refer to the following table and write latex code to create this table.

Table 1: Your first Table

Value 1 α ,	Value 2 β	Value 3 π
1	1110.10	a
2	10.1	b
3	23.113231	c

5. a) Given a data set D = {o_1, . . . , o_n} with known class labels C(o_i) ∈ C = {A, B, C} of the objects. In order to evaluate the quality of a classifier K, each object o_i ∈ D is additionally classified using K, yielding class label K(o_i). The results are given in the table below. 6+4 CO1 CO2 CO3

i	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14
C(o_i)	A	B	A	C	C	B	A	A	A	B	B	C	C	C	B
K(o_i)	A	A	C	C	B	B	A	A	A	C	A	A	C	C	B

Calculate the following:

(i) Setup the confusion matrix

(ii) Compute the accuracy

(iii) Classification error (Recall/Precision)

b) Write the short notes on:

(i) Impact factor of Journals

(ii) Databases for Computer Science discipline

(iii) Detection of Plagiarism in the report

(iv) Plagiarism and Self-Plagiarism

**********Best_of_Luck**********

Name of the Student...Riya..... Scholar Number.2K22/CSI/113

MAULANA AZAD NATIONAL INSTITUTE OF TECHNOLOGY BHOPAL
DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
EXAMINATION: MIDTERM **MONTH and YEAR: Feb 2025**

Course: M. Tech **Semester: II** **Branch: CN/IS**
Subject Code: FIS-521 **Subject Name: Privacy and Database Security**
Maximum Marks: 40 marks
Duration: 60 minutes **Date:** 18-03-2025 **Time:** 11:30 – 12:30 am

Note: 1. Attempt all questions, **2.** Assume necessary data if required and mention it.

Q. No.	Questions	Marks
1	Consider a university database system, "UniDB," that contains tables for Students, Courses, Grades, Financial_Aid, and Faculty. This database stores sensitive information including personal identification data, academic records, and financial details. For each of the CIA (Confidentiality, Integrity, Availability) triad components: a. Explain how the CAI concept applies specifically to the UniDB database. b. Identify a potential threat to that component in the context of UniDB. c. Propose a specific security control/access control that would address that threat. d. Explain how security procedures must be implemented in the above model.	(10)
2	A hospital wants to share patient data with researchers while maintaining patient privacy. Design a comprehensive privacy-preserving data mining approach that addresses technical and procedural safeguards. Justify your design choices by citing an example.	(10)
3	Compare and contrast the Bell-LaPadula Model and the Biba Integrity Model in terms of their: a) Primary security objectives b) Access control rules c) Security properties d) Limitations in practical applications	(10)
4.	Implement a Role-Based Access Control (RBAC) scheme for a hospital's patient record management system that contains patient medical histories, billing information, test results, and treatment plans. Your implementation should: a) Define appropriate roles for hospital staff (e.g., physicians, nurses, lab technicians, billing department, administrative staff) b) Establish role hierarchies that reflect the organizational structure of the hospital and clinical workflows c) Implement separation of duties to prevent conflicts of interest and reduce the risk of fraud or accidental data exposure d) Explain how your design addresses potential security vulnerabilities specific to healthcare data, including compliance with relevant healthcare privacy regulations	(10)

**********END**********

TA 2114

Name of the Student..... Scholar Number 23212011103

MAULANA AZAD NATIONAL INSTITUTE OF TECHNOLOGY BHOPAL

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

EXAMINATION: MID TERM MONTH and YEAR: MARCH 2024

Course: M.Tech Semester: II Branch: CSE Subject Code: CSE 522

Subject Name: Stochastic Process and Queuing Theory

Duration: 1 Hour 30 minutes Date: 20/03/2024 Time: 9.30 AM to 11.00 PM

Note: All questions are compulsory

Q. No.	Questions	Marks	COs
Q1.	(a) A deck of cards contains 20 cards: 6 red cards and 14 black cards. 5 cards are drawn randomly without replacement. What is the probability that exactly 4 red cards are drawn?	2	1
	(b) An oil company conducts a geological study that indicates that an exploratory oil well should have a 20% chance of striking oil. What is the probability that the first strike comes on the third well drilled?	2	1
Q2.	(a) Let X be a continuous random variable with PDF $f(x) = \{ 4x^3 \quad 0<x\leq 1$, otherwise $0\}$ Find $P(X \leq 2/3 \mid X > 1/3)$?	2	1
	(b) Three identical boxes contain red and white balls. The first box contains 3 red and 2 white balls, the second box has 4 red and 5 white balls, and the third box has 2 red and 4 white balls. A box is chosen very randomly and a ball is drawn from it. If the ball that is drawn out is red, what will be the probability that the second box is chosen?	2	1
Q3.	(a) Consider two continuous random variables X and Y with joint p.d.f. $f(x,y) = \{ (2/81)\, x^2y , 0 < x < k, 0 < y < k$, otherwise $0\}$ find the value of K and $P(x > 3y)$?	2	1 1
	(b) Suppose X is a discrete random variable and has the MGF $m_X(t) = (1/7)\, e^{2t} + (3/7)\, e^{3t} + (2/7)\, e^{5t} + (1/7)\, e^{8t}$. Find E[X] and Variance?	2	
Q4.	(a) Chocolate Company produces a Tasty Bar chocolate. The weight X of Tasty Bars is believed to be normally distributed with mean 80.3 g and standard deviation 1.3 g. What is the probability that a Tasty Bar selected at random has a weight between 79.6 g and 81.5 g? Give your answer to 3 significant figures	2	2
	(b) Derive the expression for chapman Kolmogorov equation for discrete time stochastic process.	2	2
Q5.	(a) In a town of 20000 peoples, 44% supports on upcoming referendum vote. If you are conducting a pre vote poll of the eligible voters in the town and survey will show that the referendum pass. In order for a referendum to pass it requires a majority voter or greater equal to 51%. So, this is same as asking to sum of all voters greater reached the majority line. what is the probability of getting a sample survey of 100 that shows support for the referendum.	2	3
	(b) An unknown distribution has a mean of 90 and a standard deviation of 15. A sample of size 80 is drawn randomly from the population. (i)Find the probability that the sum of the 80 values (or the total of the 80 values) is more than 7,500. (ii)Find the sum that is 1.5 standard deviations above the mean of the sums.	2	3

**********END**********

Name of the Student..... Scholar Number.....

MAULANA AZAD NATIONAL INSTITUTE OF TECHNOLOGY BHOPAL

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

EXAMINATION: MID TERM MONTH and YEAR: March 2025

Course: M.Tech Semester: II Branch: Computer Networking Subject Code: FCN 521

Subject Name: Stochastic Process and Queuing Theory

Duration: 1 Hour marks: 20 Date: 20/03/2025 Time: 11.30 AM to 12.30 PM

Note: All questions are compulsory

Q. No.	Questions	Marks	COs
Q1.	Out of every 100 jobs received at a server, 50 are of class 1, 30 of class 2 and 20 of class 3. A sample of 30 jobs is taken with replacement. (a) Find the probability that sample will contain 10 jobs of each class. (b) Find the probability that there will be exactly 12 jobs of class 2.	4	1
Q2.	(a) There are 200000 tv's and that the chance of any one of them to be out of service is 0.0025. Determine the probability that exactly three machines will be out of service on the same day. (b) An oil company conducts a geological study that indicates that an exploratory oil well should have a 20% chance of striking oil. What is the probability that the first strike comes on the third well drilled?	4	1
Q3.	(a)Arthur and Henry are rolling a fair six-sided die and the winner of their game will be the first person to get a "six". Arthur rolls the die first. Determine the probability that Arthur wins on his second throw or third throw. (b) Derive expressions for Variance of Binomial discrete probability distribution.	4	2
Q4.	(a) Let X be a continuous random variable with PDF given by $f(x)=(1/2)e^{-|x|}$ for all $x \in$ Real Number If $Y=X^2$, find the CDF of Y. (b) Let X be a continuous random variable with PDF given by $f(x)=cx^2$ for all $\|x\|<1$ Find the value of c.	4	3
Q5.	The heights of 18-year-old men are approximately normally distributed with mean 68 inches and standard deviation 3 inches. (a) What is the probability that a randomly selected 18-year-old man is between 67 and 69 inches tall. (b) If a random sample of nine 18-year-oldmen is selected, what is the probability that the mean height x is between 67 and 69 inches? (c) Is the probability in (b) higher? Why would you expect this?	4	3

**********END**********

D321201103

Name of the Student..... Scholar Number.....

MAULANA AZAD NATIONAL INSTITUTE OF TECHNOLOGY BHOPAL
DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

EXAMINATION: END TERM MONTH and YEAR: MAY 2024

Course: M.Tech Semester: II Branch: Computer Networking Subject Code: CN 522

Subject Name: Stochastic Process and Queuing Theory Max Marks: 50

Duration: 3 Hour Date: 01/05/2024 Time: 11.30 AM to 2.30 PM

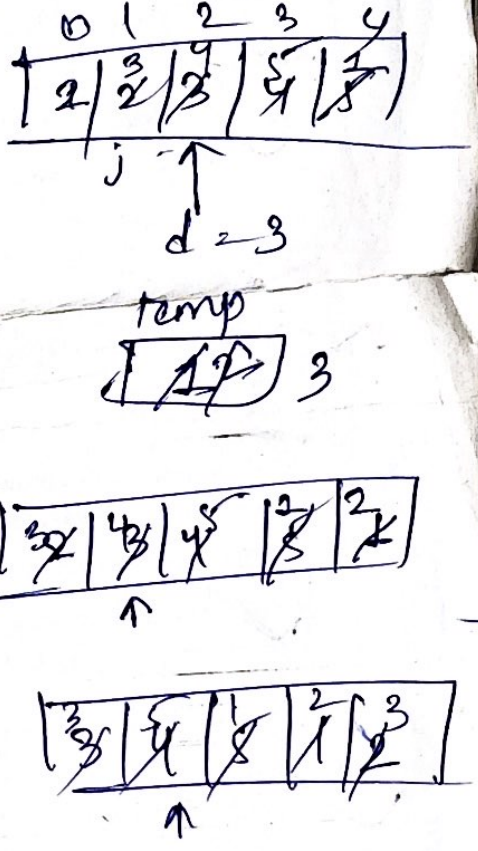
Note: All questions are compulsory

Q. No	Questions	Marks	COs
Q1.	(a) When throwing a normal die, Let assume X is the random variable and defined by X = is the square of the Score shows on the die. Then Find out the expectation of the X.	3	1
	(b) The scoring of modern IQ tests is such that Intelligence Quotients (IQs) have Normal distribution with mean 100 and variance 15. (i) What percent of people have IQ less than 80? (ii) What percent of people have IQ greater than 120? (iii) Mensa International is a non-profit organization that accepts only people with IQ within the top 2%. What level of IQ qualifies one to be a member of Mensa?	3	
	(c) Justify the statement with mathematical proof "The binomial distribution tends toward the Poisson distribution as $n \to \infty$, $p \to 0$ and np having some finite value."	4	
Q2.	(a) Three persons A, B and C have applied for a job in a private company. The chance of their selections is in the ratio 1 : 2 : 4. The probabilities that A, B and C can introduce changes to improve the profits of the company are 0.8, 0.5 and 0.3, respectively. If the change does not take place, find the probability that it is due to the appointment of C.	3	1
	(b) Derive the expressions for Total Probability theorem with the help of suitable examples.	3	
	(c) Let the joint probability density function for (X, Y) be $f(x, y) = (x+y)/3$, for $0 < x < 2, 0 < y < 1$, zero otherwise. Find the probability, $P(X > Y)$.	4	
Q3.	(a) A bank teller serves customers standing in the queue one by one. Suppose that the service time X_i for customer I has mean $E[X_i]=2$ (minutes) and $Var(X_i)=1$. We assume that service times for different bank customers are independent. Let Y be the total time the bank teller spends serving 50 customers. Find $P(90 < Y < 110)$.	4	2
	(b) Derive the expression for the var[S_n]. Where S_n is the sum of iidrv's i.e. $x_1, x_2, \ldots\ldots x_n$	3	
	(c) Derive an expression of Chapman Kolmogorov equation and master equation for transition probability of stationary and non-stationary processes.	3	
Q4.	(a) Consider the Markov chain with three states, S = {1,2,3}, that has the following transition matrix P = [{0.5, 0.25,0.25},{0.33,0,0.67},{0.5,0.5,0}]. (i) Draw the state transition diagram for this chain. (ii) If we know $P(X_1=1) = P(X1=2) = 1/4$, find $P(X_1=3, X_2=2, X_3=1)$.	4	2
	(b) Derive the Expression of P_n and P_0 for Birth and Death Stochastic Process.	3	

	(d) X is a discrete random variable, with PMF $P_x(k) = 1/3$ at $k=1$, $P_x(k) = 2/3$ at $k=2$. Find Moment Generating Function MGF of X and Its Variance.	3	3
Q5.	(a) Derive an expression of Ls and Lq for the M/M/1:/N/∞ queuing model.	4	
	(b) In railways marshalling yard, good train arrive at the rate of 30 trains per day. Assume that the inter-arrival time follows an exponential distribution and the service time is also to be assumed as exponential with mean of 30 minutes. Calculate (i) The mean queue size. (ii) The Probability that queue size exceeds to 10. (iii) If the input of the train increases to an average of 33 per day, what will be the changes in (i) and (ii).		
	(c) Explain the following Balking, Reneging and Jockeying in the context of Queuing theory.	3	

**********END**********

```
int i, j, min = 0;
for (int i=0; i<d; j++)
{
    temp = a[min];
for (j = 1; j<n; j++)
    {
        a[j-1] = a[j];
    }
    a[j] = temp;
}
```



Department of Computer Science and Engineering

Mid Term Exam (Nov - 2024)

M. Tech.(IS/CN) First Semester Time: 60 minutes Max. Marks: 20

Subject: Computer and Network Security (FIS 512/FCN 513)

Note: All questions are compulsory to attempt. Answer should be laconic. Make suitable assumptions wherever necessary and clearly state the same.

Question No	Question	CO	Marks Allotted
Q-1	A. Write the vulnerable string to exploit the following attacks. Further write defensive function/statement to handle these attacks: I. Command/Shell injection II. SQL injection	CO-2	03
	B. HLOEL WOLRD has been encrypted using a columnar transposition cipher with a key of "4312" is given: Decrypt this message using a columnar transposition cipher with a key of "4312".	CO-3	02
Q-2	A. What is non-repudiation? Can Digital Signatures be used for non-repudiation? Why?	CO-2	02

	B. Consider the following C code fragment:
	`int func(int i, double *data1, double data2)`
	`{`
	`double *p = data1;`
	`double *vec[10];`
	`if ((i<0) || (i>10)) return;`
	`vec[i] = data1;`
	`*p = data2;`
	`}`
	I. Explain why this code is vulnerable to a control hijacking attack. Briefly. explain how your attack works
	II. If this code is compiled with Stack guard, will the overflow attack be
		CO-2	03

Q-3	A. How does the base and bounds registers is used in Operating System to implement memory protection in multi-programming environment?	CO-2	04
	B. If 20MB user program reside in memory of 1GB with initial address N. Calculate the value for base and bound register.	CO-2	01

Q.4	Consider the following code snippet:
	`#include <stdio.h>`
	`#include <string.h>`
	
	`void vulnerable_function(char *input) {`
	`  char buffer[10];`
	`  strcpy(buffer, input);`
	`}`
	
	`int main(int argc, char *argv[]) {`
	`  if (argc < 2) {`
	`  printf("Usage: %s <input>\n", argv[0]);`
	`  return 1;`
	`  }`
	
	`  vulnerable_function(argv[1]);`
	`  return 0;`
	`}`		

	A. Create the Control Flow Graph (CFG) to perform static analysis.	CO-2	02
	B. Identify and analyze the vulnerabilities	CO-2	03

Subject Coordinator: Dr Deepak Singh Tomar

Name of the Student..... Scholar Number.....

MAULANA AZAD NATIONAL INSTITUTE OF TECHNOLOGY BHOPAL

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

EXAMINATION: _____ **MONTH and YEAR:** _____

Course: M-Tech **Semester:** II **Branch:** CN **Subject Code:** CN-552

Subject Name: Cloud Computing **Maximum Marks:** 20

Duration: 90 minutes **Date:** 21-03-2024 **Time:** 3:30 pm to 5:00 pm

Note: ***Each question is compulsory.***

Q. No.	Questions	Marks	COs
1	1. Which of the following is the correct statement? a) Cloud computing presents new opportunities to users and developers b) Service Level Agreements (SLAs) is small aspect of cloud computing c) Cloud computing does not have impact on software licensing d) All of the mentioned 2. What is the correct formula to calculate the cost of a cloud computing deployment? a) CostCLOUD = Σ(UnitCostCLOUD / (Revenue + CostCLOUD)) b) CostCLOUD = Σ(UnitCostCLOUD / (Revenue – CostCLOUD)) c) CostCLOUD = Σ(UnitCostCLOUD x (Revenue – CostCLOUD)) d) None of the mentioned 3. Which of the following is the Cloud Platform provided by Amazon? a) AWS b) Cloudera c) Azure d) All of the mentioned 4. Which of the following is the most refined and restrictive cloud service model? a) PaaS b) IaaS c) SaaS d) CaaS	4	CO1 ,2

2	Explain Public, Private and Hybrid cloud computing with merits and demerits.	4	CO1
3	What are the Stack of Layers from hardware appliances to software systems?	4	CO2
4	Write a short note on IaaS (Infrastructure-as-a-Service).	4	CO2

**********END**********

Name of the Student..... Scholar Number.....

MAULANA AZAD NATIONAL INSTITUTE OF TECHNOLOGY BHOPAL
DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
EXAMINATION: END-TERM May 2024

Course: M. Tech **Semester:** II **Branch:** CN
Subject Code: CN 553 **Subject Name:** Ad Hoc and Sensor NW **Duration:** 3 Hours
Date: 04-05-2024 **Maximum Marks: 60**

Note: 1. Attempt all questions, **2.** Assume necessary data, if required, and mention it.

Q. No.	Questions	Marks	COs
1 (a)	How the Ad hoc networks differ from the Sensor networks on the following characteristics: (i) Topology (ii) Routing protocols (iii) Scalability (iv) Energy resources (v) Bandwidth requirements	(3)	CO1
(b)	In an infrastructure-less Ad Hoc network, each node consumes 1 mW of power to transmit a message and 0.5 mW to receive a message. If each node sends messages to all neighbours every 40 seconds, calculate the total power consumption of the network per hour for 12 nodes.	(3)	CO1
2 (a)	What are the characteristics of wireless communication that will affect the efficiency of Ad hoc networks?	(3)	CO2
(b)	The average propagation delay between adjacent nodes is 5 milliseconds. If a packet needs to traverse 4 hops to reach its destination, calculate the end-to-end delay the packet experiences.	(3)	CO2
3 (a)	Explain the operation of routing protocols in Ad Hoc networks. Discuss different types of proactive, reactive, and hybrid routing protocols.	(3)	CO2
(b)	In an Ad Hoc network with 20 nodes, each node periodically broadcasts routing update messages to all neighbours every 30 seconds. If each routing update message is 100 bytes in size and the network operates on a single channel, calculate the total routing overhead generated in the network per hour.	(3)	CO2
4 (a)	Describe the characteristics of sensor networks and their key differences from Ad hoc networks and VANET.	(3)	CO3
(b)	In a sensor network deployment, 50 sensor nodes are distributed over 1000 square meters. Each sensor node has a transmission range of 50 meters and operates at a transmission power of 10 mW. Due to interference and obstacles, the received signal strength at a distance of 25 meters from the transmitter is attenuated to 50% of its original strength. (i) Calculate the received power at a node located 25 meters away from a transmitting node (ii) Determine whether the received signal strength is sufficient for reliable communication (assuming received power of more than 40 mW is considered sufficient) and no other sources of attenuation/noise.	(3)	CO3

5 (a)	Describe VANETs and their applications in intelligent transportation networks. What are the key benefits of VANETs in intelligent transportation networks?	(3)	CO3
(b)	What are the security and privacy concerns in VANETs? Discuss potential threats and vulnerabilities and propose strategies to mitigate security and privacy in vehicular networks.	(3)	CO3
6 (a)	Explain the communication protocols used in VANETs. Describe Dedicated Short Range Communication (DSRC) and Wireless Access in Vehicular Environments (WAVE) in detail.	(3)	CO3
(b)	In a VANET deployment, vehicles are equipped with On-Board Units (OBUs) operating in the IEEE 802.11p standard at a frequency of 5.9 GHz. Each OBU's communication range is 300 meters. Assume a highway scenario where vehicles travel at an average speed of 100 km/h. If the average packet size is 500 bytes and the data rate is 6 Mbps, calculate the time taken to transmit a single packet between two vehicles.	(3)	CO3

**********END**********

(Dr Vasudev Dehalwar)

Name of the Student..... Scholar Number.....

MAULANA AZAD NATIONAL INSTITUTE OF TECHNOLOGY BHOPAL
DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
EXAMINATION: MID-TERM

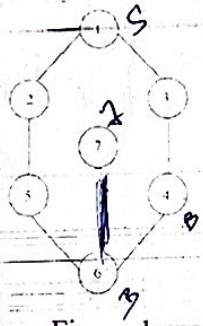
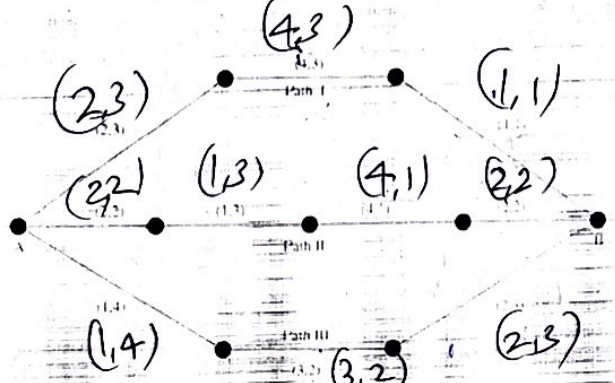
March 2024

Course: M. Tech **Semester:** II **Branch:** CN

Subject Code: CN 553 **Subject Name:** Network & System Securities

Duration: 3 Hours **Date:** 22-03-2024 **Maximum Marks: 20**

Note: 1. Attempt all questions, **2.** Assume necessary data, if required, and mention it.

Q. No.	Questions	Marks	COs
1	Differentiate between MANETs and VANETs. Discuss the key constraints and challenges commonly faced with each type of network.	(4)	CO1
2	What are the core functions of routing? Explain the difference between Ad hoc On-demand distance vector routing and Dynamic source routing.	(4)	CO2
3	What are the parameters used to define clustering in VANET? Explain any two active and passive clustering.	(4)	CO3
4	For the topology shown in Figure 1, create a DAG for node 1 as the source and node 7 as the destination in TORA. If the link between nodes 4 and 6 breaks as shown in the figure, find the change in the DAG. Also. mark the distance from the destination.  Figure 1	(4)	CO2
5	A network offers three paths between nodes A and B, as shown in Figure 2. The numbers on each link indicate (energy, delay) of the link. If the cost-field approach is followed for routing, which path will be chosen if the metric employed is (a) energy, (b) delay, and (c) energy × delay?  Figure 2	(4)	CO2

**********END**********

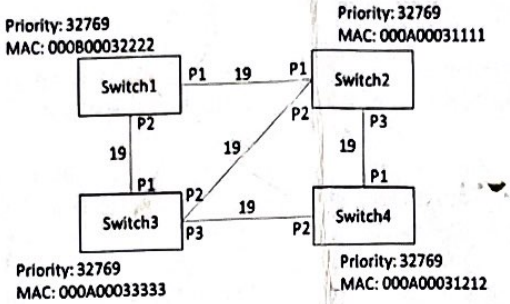
2321201103

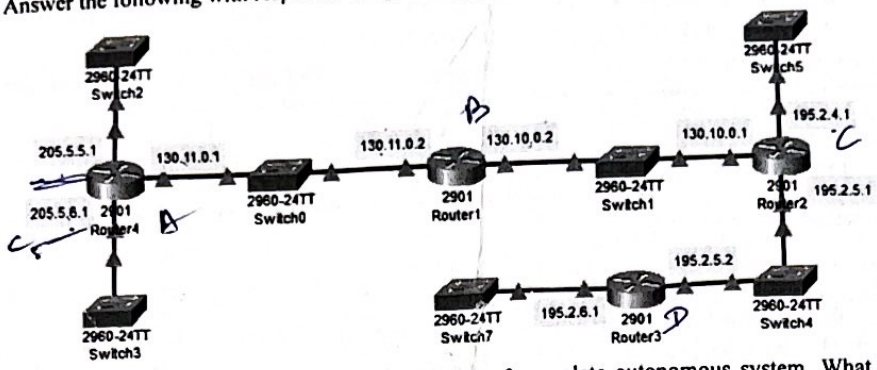
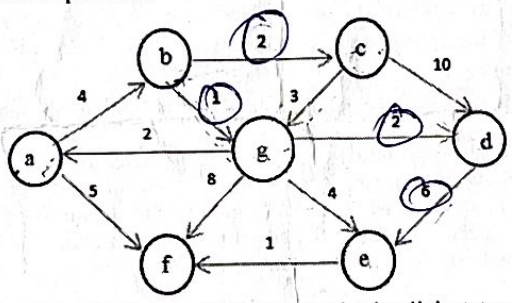
MAULANA AZAD NATIONAL INSTITUTE OF TECHNOLOGY BHOPAL
DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

EXAMINATION: Mid-Term **MONTH and YEAR: March 2024**

Course: M.Tech Semester: II **Branch: CSE (CN)** **Subject Code: CN521**
Subject Name: Network Devices and Programming **Maximum Marks: 20**
Duration: 90 MINS **Date: 19/03/2024** **Time: 9:30 AM to 11:00 AM**

Note: *Answer All Questions.*

Q. No.	Questions	Marks	COs
1	Consider the following broadcast network having redundant paths. The value on links indicates the costs. The connected ports are labelled as P1, P2, .. etc.  Answer the following with respect to the given information: a) What types of issues can be caused due to redundant links in the given network? b) How does the Spanning tree protocol address the challenges posed by the redundant links? Illustrate the working of the Spanning tree protocol in the given network. Provide suitable justifications for the choices made by the algorithm. c) Compare and contrast between the listening and learning STP state of the ports.	2+2+1	CO1
2	a. An Internet Service Provider has following chunk of CIDR-based IP addresses available: 245.248.128.0/20. The ISP wants to give first quarter of this chunk addresses to organization A, second quarter to organization B while the remaining half to organization C. What will be the following for Organization A, B and C: Subnet ID, Broadcast address, Number of IP address, Number of usable hosts per network, Usable IP address range. b. Consider the following forwarding table of a router: Subnet ID | Subnet Mask | Interface 142.17.0.0 | 255.255.0.0 | 1 142.17.64.0 | 255.255.224.0 | 2 142.17.68.0 | 255.255.255.0 | 3 142.17.68.64 | 255.255.255.224 | 4 Default | | 0 If a packet addressed to a destination address 142.17.68.117 arrives at the router, to which interface will it be forwarded? Justify your answer.	3+2	CO1
3	Consider the given network topology. Assume default subnet mask for each network and that RIP routing protocol is enabled on all the routers.	2+1+2	CO5

	Answer the following with respect to the given information:  a. Assume that Router1 has the knowledge of complete autonomous system. What is periodic RIP response sent by Router1. Provide the value of every field of RIP response packet. b. Suppose Router2 does not receive update for network 205.5.5.0, 205.5.6.0 for 240s. How many timers will be running on Router2? Provide suitable justifications in support of your answer. c. How does Split Horizon with Poisson Reverse deals with the issue of 2-node loop instability problem in distance Vector Routing? Mention the pro and con of this method over simple Split Horizon.		
4	a. Link State Routing algorithms tend to obtain shortest path tree for each node. Consider the given network topology where the nodes represent the routers and the links are labelled with the associated cost. Simulate the Dijkstra'a algorithm on Router 'b' to find the shortest path tree.  b. Information about the links in OSPF is shared using link state update packets. What all information is carried in these Link State update packets? Elaborate upon the fields of such packets with respect to Router Link LSA.	3+2	CO5

**********END**********

Name of the Student..Riya..... Scholar Number..2421201113

मौलाना आजाद राष्ट्रीय प्रौद्योगिकी संस्थान, भोपाल- 462003

MAULANA AZAD NATIONAL INSTITUTE OF TECHNOLOGY, BHOPAL- 462003

संगणक विज्ञान एवं अभियांत्रिकी विभाग

Department of Computer Science and Engineering

Mid-Term Examination, Nov 2024

Course: M.Tech (CN) **Semester:** I **Branch:** CSE

Subject: TCP-IP **Subject Code:** FCN 511

Date: 19-11-2023 **Max Marks: 20**

Time: 09:30AM to 10:30AM

Notes: Take necessary justification when required and clearly explain all the steps.

Q. No.	Question	Marks	COs
Q1.	Explain why CSMA/CD is unsuitable for wireless networks, specifically addressing the issues related to signal propagation and collision detection in a shared medium. How does CSMA/CA address these challenges, and what mechanisms are used to minimize collisions in wireless communications?	4	CO1
Q2.	A service provider has multiple Class C networks that need to be aggregated to simplify routing. Explain how Classless Inter-Domain Routing (CIDR) is used for aggregation in this scenario. Given four networks (192.168.1.0/24, 192.168.2.0/24, 192.168.3.0/24, 192.168.4.0/24), provide the supernet mask and explain the advantages of using this supernet.	4	CO1 & CO2
Q3.	The IPv4 header includes several fields that support fragmentation. Explain the roles of the identification, flags, and fragmentation offset fields in the fragmentation process. Given an IP packet of size 4000 bytes (including 20 bytes of header) being sent over a network with an MTU of 1500 bytes, demonstrate how the packet will be fragmented. Provide details of each fragment, including their sizes and header fields.	4	CO1 & CO2
Q4.	In a network using the Distance Vector Routing Protocol, discuss the "Count to Infinity" problem. Use an example with three routers (A, B, C) to demonstrate how this issue arises. Explain how split horizon and poison reverse mitigate this problem.	4	CO1 & CO2
Q5.	Prove why the Selective Repeat (SR) protocol's send window size must be less than or equal to 2^{m-1}, where m is the number of bits used for sequence numbers. Use a suitable example to illustrate your explanation.	4	CO1 & CO2

---------- END ----------

Scholar No:..... Scholar Name:.....

MAULANA AZAD NATIONAL INSTITUTE OF TECHNOLOGY, BHOPAL
DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
END TERM EXAMINATION (December 2024)

Course: M.Tech. Semester: I Date:21/12/2024
Time: 2 hour Max. Marks: 40

Sub: Web Search and Information Retrieval

Sub Code: FAC-560/FCN 558/ FIS 561/ FAI 558

Branch: All Branches

Note: Attempt All Questions.

Q. No.	Questions	Marks /CO's
1.	Given the following set of documents, construct an inverted index and compute the following for the term "retrieval": I. **Collection frequency (CF)**. II. List of documents containing the term along with **term frequency (TF)** in each document. III. **Positional information** for the term in the documents. IV. Using **proximity operators**: a. Identify in which document the term **"retrieval" OW/3 "information"** appears. b. Identify in which document the term **"retrieval" UW/2 "information"** appears. **Note:** Perform only **tokenization** as preprocessing. Do not apply lemmatization or stop word removal. **Documents:** **Doc 1:** "Information retrieval provides powerful results." **Doc 2:** "Information systems rely on retrieval and indexing." **Doc 3:** "An inverted index aids information for retrieval."	**05** [CO1]
2.	Consider the Boolean query **mercy AND worser**. The postings list of the two words are shown. The list for mercy is augmented with skip pointers: **mercy → [4, 6, 10, 16, 25, 40, 61, 69, 76, 82, 85, 90, 93, 98, 101, 116]** **worser → [4, 6, 90, 96, 100, 101, 102]** Answer the following questions: I. How many comparisons are required if neither of the lists had skip pointers? List the comparisons sequentially. II. How many comparisons are required when Mercy has skips? List the comparisons sequentially. III. What is a common heuristic for deciding the number of skip pointers, given that the size of the postings list is P?	**05** [CO2]
3.	A search engine indexes a collection of **1 million documents**, each with an average of **500 words**. To efficiently build and store the index, the following indexing and compression techniques are considered: i. Assume the term frequency follows **Zipf's Law**, where the frequency of a term is inversely proportional to its rank. If the most frequent term appears **10 million** times, estimate how many times the **2nd, 10th, and 100th** ranked terms will occur. ii. Given that the total number of unique terms V in a collection is estimated by **Heap's Law** ($V=K.N^b$). Where N is the total number of terms, k =30, and b =0.5.	**05** [CO3]

	Estimate the size of the vocabulary for the given collection (1 million documents, average 500 words). iii. Explain how the **Blocked Sort-Based Indexing** method works for this dataset. Given that the system memory can only hold **100,000** term occurrences at a time, calculate the number of blocks needed to process the collection in a single pass.	
4	The system maintains an inverted index and utilizes **B-Trees and Reverse B-Trees** for efficient wildcard query processing. Consider the following two wildcard queries: **Query 1:** Jayasur * a **Query 2:** Sana* AND Jayasur* Describe how you would process the wildcard query. Additionally, explain the difference in processing.	**05** [CO3]
5.	The Soundex algorithm is used to encode words into phonetic representations based on their pronunciation. Consider the following list of names: **Catherine, Miller, Kathryn, Malar** i. Calculate the Soundex code for each name in the list. ii. Find the Soundex matches for the following names: **Query 1:** Kathleen **Query 2:** Mohler	**05** [CO4]
6.	Consider the task of spelling correction for an isolated word using the Levenshtein Edit Distance algorithm. The word "**LEVENSHTEIN**" has been misspelled as "**MEILENSTEIN**". Determine the minimal set of edit operations (insertions, deletions, substitutions) required to correct the misspelling.	**05** [CO4]
7.	Given the following corpus of documents and a query term "data", calculate the **Term Frequency-Inverse Document Frequency (TF-IDF)** score for the term "data" in each document. N=3 Corpus: **Document 1:** "Data science is a field that uses scientific methods, processes, algorithms, and systems to extract knowledge and insights from structured and unstructured data." **Document 2:** "Data analysis involves inspecting, cleaning, transforming, and modeling data to discover useful information." **Document 3:** "Big data technologies enable the analysis of large and complex data sets."	**05** [CO5]
8.	Given the following term frequency table for a set of documents and a query term. Assume that the log frequency is 1 for all terms, and normalization must be performed on the term frequency vector (using Euclidean normalization). The term frequency table is as follows:	**05** [CO5]

Term	Doc 1	Doc2	Doc 3	Query
Data	3	2	1	2
Science	2	0	0	1
Analysis	1	2	0	0
Knowledge	0	1	2	1
Algorithms	1	0	1	0
Systems	0	1	0	1

i. Calculate the cosine similarity between the query and each document.
ii. Determine the order of document relevancy based on the rank.

***Subject Coordinator: Dr. Muktesh Gupta**

Name of the Student..... Scholar Number.....

MAULANA AZAD NATIONAL INSTITUTE OF TECHNOLOGY BHOPAL
CENTER OF ARTIFICIAL INTELLIGENCE
MIDTERM EXAMINATION, Nov. 2024

Course: M. Tech. **Semester: I**
Subject Code: AI24512/AC24552/CN24552/IS24552
Subject Name: Machine Learning **Maximum Marks: 20**
Duration: 1 Hour **Date: 22/11/2024** **Time: 9:30 am to 10:30 am**

Q. No.	Questions	Marks
1	Design a decision Tree classifier by using the ID3 algorithm for training dataset given below. A1 and A2 are Input attributes and Target shows the output for any instance.	06

A1	M	N	N	Q	M
A2	X	X	Y	X	Y
Target	C1	C1	C2	C2	C2

2	Design a Naïve Bayes classifier for the following data. Find the prediction for the instance with F1='B'; F2='P'; F3='S', Is this instance can be classified or not? if not what modification we need to do? .								06

F1	A	A	B	C	C	B	A	B	C
F2	P	P	Q	Q	Q	P	Q	Q	P
F3	R	S	R	S	R	R	S	S	S
Target	C1	C2	C2	C2	C1	C2	C2	C1	C2

3	(A) Derive Lagrangian dual of Hard margin Support Vector Machine as an optimization problem.	(4)
	(B) What are the key parameters that influence the performance and effectiveness of the Gradient Descent algorithm? Explain their impact.	(4)

Name of the Student...Riya..... Scholar Number..24.2.2011113

INDIAN INSTITUTE OF INFORMATION TECHNOLOGY, BHOPAL

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

EXAMINATION: MID TERM **MONTH and YEAR: Nov 2024**

Course: M. Tech **Semester:** I **Branch:** CN

Subject Code: FCN-512 **Subject Name:** Wireless Networks

Maximum Marks: 40 marks

Duration: 60 minutes **Date:** 19-11-2024 **Time:** 09:30 – 10:30 am

Note: 1. Attempt all questions, **2.** Assume necessary data if required and mention it clearly.

Q. No.	Questions	Marks
1	What are the different technologies used in 1G/2G/3G/4G/5G and 6G? What are the approximate data rates and modulation techniques used in these technologies?	(8)
2	Consider a wireless communication system operating at a frequency of 2.4 GHz. The transmitter power P_t is 20 dBm, the gains of the transmitting and receiving antennas are G_t=10 dB and G_r=5 dB, respectively. The receiver power P_r is measured to be −50 dBm. Assuming free-space propagation, calculate: (i) path loss PL in dB and (ii) the distance d between the transmitter and receiver	(8)
3	The cellular system is having total available spectrum of 25 MHz and channel bandwidth per user is 25 kHz for 7-cell reuse pattern. Other parameters are: 1. Number of cells in the network: 210 2. Call duration per user: 120 seconds 3. Handoff dwell time: 2 seconds 4. Probability of call interference during a handoff: 0.02 (i) Find the total system capacity (ii) Determine the number of channels available per cell. (iii) Estimate the average interference duration per call if 5% calls experience interference. (iv) Determine the percentage of time spent on handoff during a call.	(8)
4.	What are channel assignment strategies? Explain AMPS Dupoly channel assignment strategy for minimizing adjacent channel interference.	(8)
5.	Write short notes of two:- (i) Traffic engineering (ii) Mobile radio propagation characteristics (iii) Hidden terminal and exposed terminal problem in WLAN (iv) Comparison of narrow spectrum and spread spectrum	(8)

**********END**********

35/40 ≈ 20 Rounded = 17.5

Name of the Student..Riya Kymbhare Scholar Number..2721201113

MAULANA AZAD NATIONAL INSTITUTE OF TECHNOLOGY BHOPAL

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

END TERM EXAMINATION, APRIL-MAY 2024

Course: M. Tech. **Semester: I** **Subject Name: Machine Learning**

Subject Code: AI24512/AC24552/CN24552/IS24552 **Maximum Marks: 40**

Duration: 2 Hrs. **Date: 20/12/2024** **Time: 9:30 AM to 11:30PM**

<table>
<tr><th>Q. No.</th><th>Questions</th><th>Marks</th><th>COs</th></tr>
<tr><td>1</td><td>A) What are the challenges of designing a Machine Learning System and How we can overcome to these problems?

B) Suppose we have a binary classification problem training dataset as mentioned below in the table:

<table>
<tr><td>F1</td><td>6</td><td>7</td><td>8</td><td>10</td><td>11</td><td>2</td></tr>
<tr><td>Output</td><td>0</td><td>0</td><td>0</td><td>1</td><td>1</td><td>1</td></tr>
</table>

Use the Logistic regression to find the best fit line using the Gradient Descent algorithm and calculate the updated weights and cost for 2 iterations, considering that the initial weights are $W_0 = 1$, $W_1 = 0$ and learning rate is 0.1.</td><td>04

06</td><td>CO1

CO1</td></tr>
<tr><td>2</td><td>A) Design a decision Tree by using the CART algorithm for training dataset given below. F1 and F2 are Features and Target shows the output for any instance.

<table>
<tr><td>F1</td><td>P</td><td>Q</td><td>Q</td><td>R</td><td>P</td></tr>
<tr><td>F2</td><td>X</td><td>X</td><td>Y</td><td>X</td><td>Y</td></tr>
<tr><td>Target</td><td>T1</td><td>T1</td><td>T2</td><td>T3</td><td>T2</td></tr>
</table>

Evaluate the efficiency, Precision, Recall and F1 score using the final tree for the training dataset.

B) Perform the simple Linear Regression derivation by using the Gradient Descent algorithm.</td><td>06

04</td><td>CO1

CO2</td></tr>
</table>

3	**(A)** Suppose two classification models have been developed for the same problem and both are giving similar accuracy. Discusses the scenarios of correct model selection based on your problems. **(B)** Find the equation of the hyperplane using Hard Margin SVM for given support Vectors. X1 and X2 are features and Y is target attribute. **X1** | **X2** | **Y** 2 | 4 | -1 -2 | 4 | -1 -2 | -4 | 1	**04** **06**	**CO2**
4	**A)** Consider the data points mentioned in the table below. Perform the Average linkage Agglomerative Hierarchical Clustering on this data and show the final Dendrogarm. Fins the number of clusters for threshold distance 5. **Objects** | 1 | 2 | 3 | 4 | 5 **X** | 2 | 4 | 4 | 10 | 13 **Y** | 6 | 4 | 10 | 12 | 10 **(B)** Develop a Neural Network for: (x1 NOR x2), here x1 and x2 are Binary features. Initial weights $w_1 = 0.6$, $w_2 = 0.6$, $b = -0.4$ and learning rate=0.5.	**05** **05**	**CO2**

Name of the Student..... Scholar Number.....

मौलाना आजाद राष्ट्रीय प्रौद्योगिकी संस्थान, भोपाल- 462003
MAULANA AZAD NATIONAL INSTITUTE OF TECHNOLOGY, BHOPAL- 462003
संगणक विज्ञान एवं अभियांत्रिकी विभाग
Department of Computer Science and Engineering

End-Term Examination, Dec 2024

Course: M.Tech (CN) **Semester:** I **Branch:** CSE
Subject: TCP-IP **Subject Code:** FCN 511
Date: 19-12-2024 **Max Marks:** 40
Time: 09:30AM to 11:30AM

Notes: Take necessary justification when required and clearly explain all the steps.

Q. No.	Question	Marks	COs
Q1.	Assume the ethernet LAN is configured with CSMA/CD using bus topology. If the stations are using frames of size 500 bits and the datarate offered by a cable is 50 mbps (megabits per second). The signal speed inside the cable is 2,50,000 km/sec. The stations are connected uniformly with a spacing of 10 mts between the adjacent ones. Calculate the maximum number stations that can be connected?	5	CO1
Q2.	In a network using the Distance Vector Routing Protocol, discuss the "Count to Infinity" problem. Use an example with three routers (A, B, C) to demonstrate how this issue arises. Explain how split horizon and poison reverse mitigate this problem.	5	CO1 & CO2
Q3.	For reliable frame transmission, a sender uses Stop-and-Wait ARQ protocol. Frames are 500 bytes in size, and the sender's transmission rate is 40 Kbps (1Kbps = 1000 bits per second). The size of an acknowledgement is 50 bytes, and transmission rate of the receiver is 2 Kbps. The propagation delay of the channel is 100 milliseconds. Calculate the throughput assuming no frames are lost.	5	CO1 & CO2
Q4.	Consider an IPv4 packet with option field(s) of 20 bytes. Calculate the checksum field for the IPv4 header when *service type* field is 0, *total length* field is 60, *identification* field is 1, *do not fragment* (*D*) bits is set to 1, *TTL* is 4, the higher layer protocol is TCP, initial value of the checksum is set to 0 at the sender, the *source address* is 5.11.7.5 and the *destination address* is 10.10.8.4.	5	CO1 & CO2
Q5.	Prove why the Selective Repeat (SR) protocol's send window size must be less than or equal to 2^{m-1}, where m is the number of bits used for sequence numbers. Use a suitable example to illustrate your explanation.	5	CO1 & CO2
Q6.	What challenges in IP communication require mechanisms for error reporting and diagnostics? How does ICMP address these challenges? Focus specifically on the functionality of the Destination Unreachable and Time Exceeded ICMP messages. Provide examples of scenarios where these messages are generated and analyze their importance in ensuring reliable packet delivery and effective troubleshooting in IP networks.	5	CO1 & CO2
Q7.	Describe the structure of an SCTP packet, including the roles of common control and data chunks. Provide a detailed explanation of SCTP's four-way handshake process for establishing a connection and its mechanism for handling association termination.	5	CO3
Q8.	Explain the primary purpose of the Dynamic Host Configuration Protocol (DHCP). Describe the four-step process of IP address allocation (Discover, Offer, Request, Acknowledge) in detail, including the role of DHCP servers and clients. Provide an example scenario where DHCP simplifies network configuration.	5	CO3

---------- END ----------

Name of the Student..... **Scholar Number.....**

INDIAN INSTITUTE OF INFORMATION TECHNOLOGY, BHOPAL

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

EXAMINATION: END TERM **MONTH and YEAR: Dec 2024**

Course: M. Tech **Semester:** I **Branch: CN**

Subject Code: FCN-512 **Subject Name: Wireless Networks**

Maximum Marks: 50 marks

Duration: 2 Hrs **Date:** 16-12-2024 **Time:** 09:30 – 11:30 am

Note: 1. Attempt all questions, **2.** Assume necessary data if required and mention it clearly.

Q. No.	Questions	Marks	
1	Describe the O-RAN architecture and the features of 5G wireless communication.	(5)	CO4
2	What is the technology used in MIMO? How does massive MIMO enhance the bandwidth and data rate in 5G communication?	(5)	CO4
3	Describe different media access techniques in wireless communication. How OFDM adapt or enhance media access techniques to meet modern requirements like ultra-low latency and high reliability in 5G/6G technologies?	(6)	CO3
4	What are the different media access control techniques used in WLAN? Describe CSMA/CA with Binary Exponential Backoff (BEB) algorithm.	(6)	CO3
5	What are the characteristics of normal and abnormal Hand off? A mobile user travels at 50 km/h through a cell with a radius of 1 km. The dwell time is defined as the time the user remains within the cell boundary. • Calculate the dwell time for the mobile user in the cell. • If a handoff requires 5 seconds to complete, determine if there is sufficient time for a seamless handoff.	(6)	CO2
6	Explain the concept of CDMA. Consider a CDMA system in which user A & B have the code cA= <1,-1,-1, 1, -1, 1> and cB= < 1, 1, -1, -1, 1, 1> respectively. (i) Show the output of the receiver if A transmits data bit 1 in fist case and & 0 next. (ii) Show the output of the receiver if A tries to read the B's data.	(6)	CO3
7.	Consider a cellular system in which there are total of 1001 radio channels available for handling traffic. Suppose the area of the cell is 6 km^2 and the area of the entire system is 2100 km^2 . a) Calculate the system capacity if the cluster size is 7.	(8)	CO1

	b) How many times would the cluster of size 4 have to be replicated in order to approximately cover the entire cellular area? c) Calculate the system capacity if the cluster size is 4. d) Does decreasing the cluster size increase the system capacity? Explain		
8	Write short notes any two: - (i) Mobile radio propagation models (ii) Cognitive Radio (iii) Methods to increase capacity in Cellular Networks (iv) Hidden and Exposed terminal in WLAN	8	CO2 & 3

**********END**********